

Row–Diagonal Closure at the Moving Möbius Gate: Folded Alias Energies, Exact Four–Correlation Isolation, and Triple–Prime Progression Decorrelation

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Abstract

TPC–39 isolates a finite additive–alias energy whose terminal expansion appears to contain both long–gap two–Möbius correlations and cross–row four–Möbius correlations. We prove that the first class is not an arithmetic gate. For an arbitrary unimodular alias bank with normalized Gram H , the complete same–row sector is bounded by

$$\mathcal{E}_{\text{same}} \leq \lambda_{\max}(H)\mathcal{D}, \quad |\mathcal{E}_{\text{same}} - \mathcal{D}| \leq \|H - I\|_{\text{op}}\mathcal{D},$$

where $\mathcal{D}$ is the atomic diagonal. The minimal $q = L + 1$ Fourier bank has spectrum $2, 1, 0$, hence its same–row sector is at most $2\mathcal{D}$ and the absolute value of its complete signed off–alias two–Möbius sector is at most $\mathcal{D}$. The full $2L + 1$ -point bank deletes that sector exactly. Since TPC–39 already bounds $\mathcal{D}$ at the physical endpoint, every same–row alias correlation is thereby closed at the required scale without any Möbius cancellation.

After this deletion the only unbounded sector inside the chosen minimal folded terminal energy is an explicit cross–row form. Its summands contain

$$\mu(d_\alpha)\mu(d_\beta)\mu(m_\alpha j + h + kM)\mu(m_\beta j + h + k'M), \quad \alpha \neq \beta,$$

with the actual masks and source weights. We prove that the two affine target forms are nonproportional for every retained row pair, give the exact one–sided bound equivalent to closure of that folded terminal energy, and compress each fixed–alias column into a Hilbert–valued one–Möbius correlation. This identifies precisely why a scalar completion cannot control the physical row energy.

We also obtain a quantitative arithmetic specialization at the inherited CRT scale. Let $Y = X^{133/400}$, choose three disjoint admissible prime intervals of length comparable with Y , and let $M = q_1 q_2 q_3 \asymp X^{399/400}$. For every fixed $0 < \theta < 1/200$, all but a zero-density subfamily of these triple–prime moduli satisfy the raw, uncentered bound

$$\sum_{a \bmod M}^* \left| \sum_{\substack{n \leq X \\ n \equiv a \pmod{M}}} \mu(n) \right|^2 \ll \frac{X(X/M)}{(\log X)^\theta}.$$

The proof combines the almost–all–moduli theorem of Klurman–Mangerel–Teräväinen with a maximal multiplicative large sieve that removes the single pretentious character on average over the sparse triple–prime family. A bounded–variation transfer includes the smooth terminal weight $-\mu(n) \log n \Phi_0(n/X)$. This is genuine coefficient–specific logarithmic decorrelation from the scalar coherent scale, but it remains a factor $(X/M)/(\log X)^\theta$ above the atomic diagonal and does not estimate the physical cross–row four–Möbius form. No parity breach, Hardy–Littlewood asymptotic, or twin–prime conclusion is claimed.

Keywords: Möbius correlation; arithmetic progression; folded Parseval; alias Gram; multiplicative large sieve; triple-prime CRT modulus; parity boundary.

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1 Introduction

The TPC programme has progressively separated geometric losses from the arithmetic cancellation that would be needed at a prime-pair endpoint. TPC-34 isolates an orbit-sliced Gram operator; TPC-36 gives an exact soft projective reassembly of the physical row mask; TPC-37 and TPC-38 localize the relevant CRT face and identify its identity bucket; and TPC-39 recovers the equality alias by a minimal additive Fourier bank [10–14]. TPC-40 then proves that universal row phasing cannot decorrelate the equality mode at unit recovery cost [15].

The endpoint left by these papers has two superficially similar types of terms. If a physical row α and an alias k contribute a terminal atom

$$b_{\alpha,j,k} = -\mu(m_\alpha j + h + kM) \log(m_\alpha j + h + kM) \Phi_0(m_\alpha j + h + kM), \quad (1)$$

then expansion of the folded square produces

- (i) same-row products with $\alpha = \beta$ and $k \neq k'$, which look like a long-gap two-Möbius correlation; and
- (ii) cross-row products with $\alpha \neq \beta$, which also retain the two source factors $\mu(d_\alpha)\mu(d_\beta)$ and are therefore four-Möbius expressions.

The first purpose of this paper is to show that these are not two versions of the same analytic difficulty.

1.1 The row-diagonal closure

Let k range over an alias interval, let y denote an output coordinate, and consider measurements

$$Y_{t,y} = \sum_{\alpha,k} \theta_{t,k} c_{\alpha,y} b_{\alpha,y,k}, \quad \sum_t w_t = 1, \quad |\theta_{t,k}| = 1. \quad (2)$$

The normalized alias Gram is

$$H_{kk'} = \sum_t w_t \overline{\theta_{t,k}} \theta_{t,k'}. \quad (3)$$

If $\mathcal{D}$ is the atomic diagonal, the exact contribution from $\alpha = \beta$ obeys

$$\boxed{\mathcal{E}_{\text{same}} \leq \lambda_{\max}(H)\mathcal{D}, \quad |\mathcal{E}_{\text{same}} - \mathcal{D}| \leq \|H - I\|_{\text{op}}\mathcal{D}.} \quad (4)$$

This elementary Gram fact has a decisive endpoint consequence. For the minimal bank of TPC-39, H is a direct sum of one scalar block and two-by-two all-ones blocks. Therefore

$$\mathcal{E}_{\text{same}} \leq 2\mathcal{D}, \quad |\mathcal{E}_{\text{same}} - \mathcal{D}| \leq \mathcal{D}. \quad (5)$$

The apparently difficult two-Möbius sum is just an inner product inside each folded pair and is absorbed by $2|uv| \leq |u|^2 + |v|^2$. A full alias DFT has $H = I$ and deletes it exactly.

Since the atomic diagonal was already proved to have size

$$\mathcal{D}_L + \mathcal{D}_R \ll X^{o(1)} Q^2 J K_B = X^{o(1)} \frac{Q^3}{J}, \quad (6)$$

the entire same-row sector is closed. This removes a false arithmetic gate rather than proving a special case of Chowla's conjecture.

1.2 The remaining four–Möbius form

For the minimal bank put

$$u_{\alpha,j,0} = b_{\alpha,j,0}, \quad u_{\alpha,j,r} = b_{\alpha,j,r} + b_{\alpha,j,r-q} \quad (1 \leq r \leq L), \quad (7)$$

where $q = L + 1$. After the row diagonal is removed, the left terminal remainder is exactly

$$\mathcal{F}_L = \sum_{\gamma,j} \sum_{a=0}^L \sum_{\alpha \neq \beta} c_{\alpha,\gamma,j}^L \overline{c_{\beta,\gamma,j}^L} u_{\alpha,j,a} \overline{u_{\beta,j,a}}, \quad (8)$$

with a symmetric right form $\mathcal{F}_R$. The necessary and sufficient one–sided endpoint input, after the already proved diagonal estimate, is

$$\boxed{\operatorname{Re}(\mathcal{F}_L + \mathcal{F}_R) \ll X^{o(1)} Q^2 J K_B.} \quad (9)$$

An absolute–value estimate would be stronger than required.

Every expanded row pair contains two nonproportional affine forms in the orbit variable j . Nevertheless, their slopes and intercepts grow with X , the coefficients depend on the shared opposite row, and the target is an aggregate Hilbert norm. Existing fixed–form Chowla theorems do not provide the required uniformity. We therefore also give an exact vector compression that writes a terminal column as

$$\mathcal{T}_k^L = - \sum_r \mu(r + h + kM) \mathbf{G}_k(r), \quad (10)$$

where $\mathbf{G}_k(r)$ takes values in the physical output Hilbert space. This formulation exposes the missing theorem without expanding into four Möbius factors.

1.3 A positive scalar arithmetic step

The physical vector gate remains open, but the scalar CRT shadow admits a strict coefficient–specific saving with a sharpened exceptional–set specialization. Write

$$Y = X^{133/400}, \quad M \asymp Y^3, \quad K_M = X/M \asymp X^{1/400}. \quad (11)$$

We specialize the almost–all–moduli theorem of Klurman et al. [3] to products of three primes in disjoint admissible intervals. Its rank–one character subtraction is then removed on average over this sparse family by a maximal multiplicative large sieve. For any fixed $0 < \theta < 1/200$, almost all of the resulting moduli satisfy

$$\boxed{\sum_{a \bmod M}^* \left| \sum_{\substack{n \leq X \\ n \equiv a \pmod{M}}} \mu(n) \right|^2 \ll \frac{X K_M}{(\log X)^\theta}.} \quad (12)$$

The coefficient–blind scale is $X K_M$, so (12) is genuine arithmetic cancellation. The diagonal scale is only X , however; a factor $K_M/(\log X)^\theta$ remains. Moreover, this scalar residue–class energy contains neither the physical row synthesis nor the shared masks in (8).

The distinction is central to the claim boundary. Classical Barban–Davenport–Halberstam theory already implies logarithmic savings for almost all moduli in this scalar range. Our arithmetic contribution is the explicit admissible triple–prime specialization, its stronger exceptional–density ledger, and transfer to the terminal weight. The paper also removes the same–row alias sector from the selected folded physical gate. It does not control the cross–row four–Möbius form, cross the sieve parity barrier [1], prove a prime–pair asymptotic, or imply infinitely many twin primes.

1.4 Organization

Section 2 fixes the inherited scales and terminal packet. Sections 3 and 4 prove the general Gram identity and its minimal/full Fourier consequences. Sections 5 and 6 isolate the exact four–Möbius remainder and its Hilbert compression. Sections 7 and 8 prove the raw triple–prime progression theorem and its smooth–weight transfer. Sections 9 to 11 compare ideal convolution completion, known arithmetic inputs, and the next physical gate. The exact finite certificate is recorded in section 12.

2 Inherited physical interface

We restate only the terminal data from TPC–36 and TPC–39 needed in the new arguments. All estimates in this section are inherited; the row–diagonal theorem in the next section is abstract and independent of their derivation.

2.1 Rows and endpoint scales

Fix $h \in \mathbb{Z} \setminus \{0\}$. A retained physical row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad \ell_\alpha \asymp L_0, \quad d_\alpha \asymp D_0, \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad L_0 D_0 = Q, \quad QJ \asymp X, \quad (13)$$

where the source separation makes m_α determine α . The two source coefficient systems obey

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (14)$$

The orbit variable j ranges through $O(J)$ positive integers.

Choose three distinct auxiliary primes $q_1, q_2, q_3 \asymp J$ with

$$\max \mathcal{J} < q_i, \quad \max_\alpha d_\alpha < q_i < \min_\alpha \ell_\alpha, \quad q_i \nmid h, \quad (15)$$

and put

$$M = q_1 q_2 q_3. \quad (16)$$

The physical regular packet has $(m_\alpha j + h, M) = 1$; this property is unchanged after adding any multiple of M .

At the optimized endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad M = X^{399/400+o(1)}. \quad (17)$$

If every completed determinant satisfies $|F| \leq B \asymp X$, define

$$L = \left\lfloor \frac{B}{M} \right\rfloor, \quad q = L + 1, \quad \mathcal{I}_B = \{-L, \dots, L\}, \quad K_B = 2L + 1. \quad (18)$$

Then

$$K_B = X^{1/400+o(1)} = X^{o(1)} \frac{Q}{J^2}. \quad (19)$$

With $B_0 = Q^2 J$, the endpoint ledger is

$$\boxed{B_0 K_B = X^{o(1)} \frac{Q^3}{J} = X^{167/100+o(1)}}. \quad (20)$$

2.2 Terminal atoms and physical coefficients

For $k \in \mathcal{I}_B$, set

$$n_{\alpha,j,k} = m_\alpha j + h + kM \quad (21)$$

and, on the positive terminal support,

$$b_{\alpha,j,k} = -\mu(n_{\alpha,j,k}) \log(n_{\alpha,j,k}) \Phi_0(n_{\alpha,j,k}). \quad (22)$$

It is zero outside that support. The left terminal output at fixed alias has coordinates

$$[\mathcal{J}_k^L]_{\gamma,j} = \sum_{\alpha} c_{\alpha,\gamma,j}^L b_{\alpha,j,k}, \quad c_{\alpha,\gamma,j}^L = \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j). \quad (23)$$

The right output is obtained by interchanging the two row systems. The actual multiplier contains the row separation, gcd, residue, and smooth masks and satisfies

$$\sup_{\alpha,\gamma,j} |\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)| \ll X^{o(1)}. \quad (24)$$

In particular, the shift kM changes the terminal atom, not the inherited row–time multiplier.

The source coefficient $\gamma_{\alpha}^{(1)}$ includes the factor $\mu(d_{\alpha})$. Consequently a cross–row expansion with $\alpha \neq \beta$ contains $\mu(d_{\alpha})\mu(d_{\beta})$ in addition to the two target Möbius factors. We will retain the compact notation c^L until that expansion is needed.

2.3 The inherited diagonal theorem

Define the left and right atomic diagonals by

$$\mathcal{D}_L = \sum_{\gamma,j} \sum_{\alpha,k} |c_{\alpha,\gamma,j}^L|^2 |b_{\alpha,j,k}|^2, \quad (25)$$

$$\mathcal{D}_R = \sum_{\alpha,j} \sum_{\gamma,k} |c_{\gamma,\alpha,j}^R|^2 |b_{\gamma,j,k}|^2. \quad (26)$$

TPC–39 proves

$$\boxed{\mathcal{D}_L + \mathcal{D}_R \ll X^{o(1)} Q^2 J K_B = X^{o(1)} \frac{Q^3}{J}.} \quad (27)$$

This estimate uses the row ℓ^2 ledger and bounded physical masks; it does not use Möbius cancellation. The new point is that it controls more than the literal atomic diagonal after the alias Gram is taken into account.

Finally, TPC–39 proves terminal row–alias injectivity: for every fixed j , the map

$$(\alpha, k) \mapsto m_{\alpha} j + h + kM \quad (28)$$

is injective. This follows from $(j, M) = 1$, source separation, and

$$\max_{\alpha,\beta} |m_{\alpha} - m_{\beta}| < M, \quad (29)$$

which is automatic at (17).

3 A general row–diagonal identity

We first separate alias geometry from arithmetic. The theorem below is valid for arbitrary complex atoms and arbitrary output coordinates.

Let $\mathcal{Y}, \mathcal{R}, \mathcal{K}, \mathcal{T}$ be finite index sets. Think of $y \in \mathcal{Y}$ as an output coordinate, $\alpha \in \mathcal{R}$ as a physical row, $k \in \mathcal{K}$ as an alias, and $t \in \mathcal{T}$ as a phase measurement. Let

$$Y_{t,y} = \sum_{\alpha \in \mathcal{R}} \sum_{k \in \mathcal{K}} \theta_{t,k} c_{\alpha,y} b_{\alpha,y,k}, \quad (30)$$

where $|\theta_{t,k}| = 1$. Give the measurements weights $w_t > 0$ with $\sum_t w_t = 1$, and define

$$H_{k,k'} = \sum_{t \in \mathcal{T}} w_t \overline{\theta_{t,k}} \theta_{t,k'}, \quad (31)$$

$$\mathcal{E} = \sum_{t \in \mathcal{T}} w_t \sum_{y \in \mathcal{Y}} |Y_{t,y}|^2, \quad (32)$$

$$\mathcal{D} = \sum_{y, \alpha, k} |c_{\alpha,y}|^2 |b_{\alpha,y,k}|^2. \quad (33)$$

The matrix H is positive semidefinite and has unit diagonal.

Theorem 3.1 (Row–diagonal Gram closure). *The sector of $\mathcal{E}$ with equal physical rows is exactly*

$$\boxed{\mathcal{E}_{\text{same}} = \sum_{y, \alpha} |c_{\alpha,y}|^2 b_{\alpha,y}^* H b_{\alpha,y}}, \quad (34)$$

where $b_{\alpha,y} = (b_{\alpha,y,k})_{k \in \mathcal{K}}$. Hence

$$0 \leq \mathcal{E}_{\text{same}} \leq \lambda_{\max}(H) \mathcal{D}. \quad (35)$$

If $\mathcal{S}_{\text{same}}$ denotes the part with equal rows but unequal aliases, then

$$\mathcal{S}_{\text{same}} = \mathcal{E}_{\text{same}} - \mathcal{D} \quad (36)$$

and

$$\boxed{|\mathcal{S}_{\text{same}}| \leq \|H - I\|_{\text{op}} \mathcal{D}}. \quad (37)$$

No sign, independence, or cancellation property of the atoms is used.

Proof. Expanding (32) and averaging the phases gives

$$\mathcal{E} = \sum_y \sum_{\alpha, \beta} \sum_{k, k'} \overline{c_{\alpha,y}} c_{\beta,y} \overline{b_{\alpha,y,k}} H_{k,k'} b_{\beta,y,k}. \quad (38)$$

Restricting to $\alpha = \beta$ proves (34). Positivity of H and the Rayleigh quotient give (35). Since $H_{k,k} = 1$, the alias diagonal in (34) is precisely $\mathcal{D}$, so the remainder is (36). Finally,

$$|b^*(H - I)b| \leq \|H - I\|_{\text{op}} \|b\|_2^2$$

for each (α, y) ; summing with the nonnegative weights $|c_{\alpha,y}|^2$ proves (37). $\square$

Remark 3.2 (Why this is stronger than termwise arithmetic). If the entries of b contain Möbius values, the off–alias sector is visually a two–Möbius correlation. [Theorem 3.1](#) bounds its *whole signed quadratic form* by the diagonal. It does not assert cancellation in any individual shift correlation, and none is required for the phase energy.

Corollary 3.3 (Uniform bounded–Gram criterion). *For any family of banks with $\sup \lambda_{\max}(H) < \infty$, the same–row sector is controlled by the atomic diagonal with a uniform constant. If $H = I$, every unequal alias term with $\alpha = \beta$ vanishes identically.*

The hypothesis that the phase depends only on (t, k) is important. For a coupled phase $\theta_{t,\alpha,k}$, one obtains a separate alias Gram $H^{(\alpha)}$ for every fixed row. Exact recovery alone does not force $\lambda_{\max}(H^{(\alpha)}) = O(1)$; the recovery–coherence uncertainty of TPC–40 explains why this cannot be assumed for free.

4 Minimal folding and full Fourier deletion

We now specialize [theorem 3.1](#) to the two phase banks naturally attached to the alias interval $\mathcal{I}_B = \{-L, \dots, L\}$.

4.1 The minimal equality–recovery bank

Put $q = L + 1$, $\omega = \exp(2\pi i 1/q)$, and

$$\theta_{\nu,k} = \omega^{\nu k}, \quad 0 \leq \nu < q, \quad w_\nu = q^{-1}. \quad (39)$$

Its alias Gram is

$$H_{k,k'} = \frac{1}{q} \sum_{\nu=0}^{q-1} \omega^{\nu(k'-k)} = \mathbf{1}_{k \equiv k' \pmod{q}}. \quad (40)$$

The residue bucket 0 consists only of $\{0\}$, while bucket r , $1 \leq r \leq L$, is

$$\{r, r - q\}. \quad (41)$$

Thus H is the direct sum of $[1]$ and L copies of the two–by–two all–ones matrix.

Corollary 4.1 (Sharp minimal–bank row closure). *For the bank [\(39\)](#),*

$$\text{spec}(H) = \{2 \text{ (} L \text{ times)}, 1 \text{ (} 1 \text{ time)}, 0 \text{ (} L \text{ times)}\}. \quad (42)$$

Consequently

$$\boxed{0 \leq \mathcal{E}_{\text{same}} \leq 2\mathcal{D}, \quad |\mathcal{S}_{\text{same}}| \leq \mathcal{D}.} \quad (43)$$

Both constants are sharp in the unrestricted atom class.

Proof. The spectrum follows from the block decomposition. Hence $\lambda_{\max}(H) = 2$ and $\|H - I\|_{\text{op}} = 1$, and [theorem 3.1](#) applies. Equal atoms on one folded pair attain the first constant; opposite atoms attain the absolute value in the second. $\square$

There is also a direct form of the second inequality:

$$\mathcal{S}_{\text{same}} = 2 \operatorname{Re} \sum_{y,\alpha} \sum_{r=1}^L |c_{\alpha,y}|^2 b_{\alpha,y,r} \overline{b_{\alpha,y,r-q}}. \quad (44)$$

Every nonzero alias belongs to exactly one pair in [\(41\)](#). Applying $2|uv| \leq |u|^2 + |v|^2$ within each pair proves

$$|\mathcal{S}_{\text{same}}| \leq \sum_{y,\alpha} \sum_{k \neq 0} |c_{\alpha,y}|^2 |b_{\alpha,y,k}|^2 \leq \mathcal{D}. \quad (45)$$

This makes explicit why the gap

$$r - (r - q) = q, \quad qM \in (B, B + M], \quad (46)$$

does not create a two–point Chowla requirement.

4.2 The full alias DFT

Let $K_B = 2L + 1$, $\zeta = \exp(2\pi i / K_B)$, and instead take

$$\theta_{\nu,k} = \zeta^{\nu k}, \quad 0 \leq \nu < K_B, \quad w_\nu = K_B^{-1}. \quad (47)$$

Since distinct $k, k' \in [-L, L]$ remain distinct modulo K_B ,

$$H = I_{K_B}. \quad (48)$$

Corollary 4.2 (Exact full-DFT deletion). *For the full bank,*

$$\boxed{\mathcal{E}_{\text{same}} = \mathcal{D}, \quad \mathcal{S}_{\text{same}} = 0.} \quad (49)$$

Thus every same-row unequal-alias product is deleted before any arithmetic estimate.

More strongly, inserting $H = I$ into the full expansion gives

$$\mathcal{E}^{\text{full}} = \sum_{k=-L}^L \sum_y \left| \sum_{\alpha} c_{\alpha,y} b_{\alpha,y,k} \right|^2. \quad (50)$$

Hence the full DFT deletes every product with $k \neq k'$, including cross-row products. Its only off-diagonal terms are the fixed-alias row correlations parameterized by the Hilbert columns in [section 6](#).

The full bank uses $2L + 1$ measurements instead of $L + 1$. Since both are $\asymp K_B$, this changes no endpoint exponent. The minimal bank is optimal for recovering the equality column, while the full bank is the cleaner sufficient object if one wants to remove all alias cross terms.

4.3 Physical endpoint consequence

Apply [corollary 4.1](#) first to the left coordinates $y = (\gamma, j)$ and then to the symmetric right coordinates. Using [\(27\)](#) gives

$$\boxed{\mathcal{E}_{\text{same},L} + \mathcal{E}_{\text{same},R} \ll X^{o(1)} Q^2 J K_B = X^{o(1)} \frac{Q^3}{J}.} \quad (51)$$

In particular, the complete weighted signed sector built from the terminal products

$$\mu(n_{\alpha,j,r}) \mu(n_{\alpha,j,r-q}) \quad (52)$$

that survive minimal folding is already absorbed at the physical target scale. This does not bound any individual long-shift correlation. It is an unconditional theorem about the actual terminal packet, not a heuristic randomness assertion.

5 Exact isolation of the cross-row four-Möbius form

The row-diagonal theorem leaves one terminal sector. We now write it without hiding its signs, masks, or alias pairs.

5.1 Folded physical atoms

For the minimal bank define

$$u_{\alpha,j,0} = b_{\alpha,j,0}, \quad u_{\alpha,j,r} = b_{\alpha,j,r} + b_{\alpha,j,r-q} \quad (1 \leq r \leq L). \quad (53)$$

Hilbert-valued folded Parseval gives the exact left identity

$$\begin{aligned} \mathcal{E}_L^{\text{fold}} &:= \frac{1}{q} \sum_{\nu=0}^{q-1} \sum_{\gamma,j} \left| \sum_{\alpha,k} \omega^{\nu k} c_{\alpha,\gamma,j}^L b_{\alpha,j,k} \right|^2 \\ &= \sum_{\gamma,j} \sum_{a=0}^L \left| \sum_{\alpha} c_{\alpha,\gamma,j}^L u_{\alpha,j,a} \right|^2. \end{aligned} \quad (54)$$

Splitting the last square according to $\alpha = \beta$ and $\alpha \neq \beta$, put

$$\mathcal{E}_{\text{same},L} = \sum_{\gamma,j} \sum_{a=0}^L \sum_{\alpha} |c_{\alpha,\gamma,j}^L|^2 |u_{\alpha,j,a}|^2, \quad (55)$$

$$\mathcal{F}_L = \sum_{\gamma,j} \sum_{a=0}^L \sum_{\alpha \neq \beta} c_{\alpha,\gamma,j}^L \overline{c_{\beta,\gamma,j}^L} u_{\alpha,j,a} \overline{u_{\beta,j,a}}. \quad (56)$$

The ordered row-pair sum is real, but we retain Re below to make clear that only an upper bound is used. Define $\mathcal{E}_{\text{same},R}$ and $\mathcal{F}_R$ symmetrically.

Theorem 5.1 (Exact terminal reduction). *The minimal terminal bank satisfies*

$$\mathcal{E}_L^{\text{fold}} + \mathcal{E}_R^{\text{fold}} = \mathcal{E}_{\text{same},L} + \mathcal{E}_{\text{same},R} + \text{Re}(\mathcal{F}_L + \mathcal{F}_R). \quad (57)$$

The first two terms obey the endpoint target by (51). Consequently the terminal folded estimate

$$\mathcal{E}_L^{\text{fold}} + \mathcal{E}_R^{\text{fold}} \ll X^{o(1)} Q^2 JK_B \quad (58)$$

is equivalent, up to the already proved diagonal term, to the one-sided cross-row bound

$$\boxed{\text{Re}(\mathcal{F}_L + \mathcal{F}_R) \ll X^{o(1)} Q^2 JK_B.} \quad (59)$$

Proof. Equation (57) is the partition of the ordered double row sum in (54). The same-row part is nonnegative and has the bound (51). Hence (59) implies (58). Conversely, $\text{Re}(\mathcal{F}_L + \mathcal{F}_R)$ is the total energy minus a nonnegative same-row energy, so (58) implies the same one-sided upper bound. $\square$

This equivalence explains why estimating $|\mathcal{F}_L| + |\mathcal{F}_R|$ is more than is logically necessary. A signed dispersion argument may exploit cancellation between cross-row terms as long as it produces the upper bound in (59).

5.2 The literal arithmetic expansion

For $a = 0$, the only alias pair is $(k, k') = (0, 0)$. For $1 \leq a = r \leq L$, expansion of $u_{\alpha,j,r} \overline{u_{\beta,j,r}}$ produces

$$(k, k') \in \{(r, r), (r, r - q), (r - q, r), (r - q, r - q)\}. \quad (60)$$

The source factors inside c^L and the target atoms inside u show that every summand with $\alpha \neq \beta$ contains

$$\boxed{\mu(d_\alpha) \mu(d_\beta) \mu(m_\alpha j + h + kM) \mu(m_\beta j + h + k'M)}. \quad (61)$$

It is multiplied by the two source logarithms, the two target logarithms, the terminal cutoffs, the shared opposite-row masks, and the inherited residue and smooth factors. Formula (61) is therefore a description of the actual coefficient kernel, not a replacement by an unweighted Chowla model.

5.3 Nonproportionality of every cross-row pair

For fixed retained α, β, k, k' , define affine forms in j by

$$L_{\alpha,k}(j) = m_{\alpha}j + h + kM, \quad L_{\beta,k'}(j) = m_{\beta}j + h + k'M. \quad (62)$$

Lemma 5.2 (Uniform algebraic nondegeneracy). *If $\alpha \neq \beta$, the two forms in (62) are nonproportional for every retained k, k' .*

Proof. Proportionality would force their coefficient determinant to vanish:

$$m_{\alpha}(h + k'M) = m_{\beta}(h + kM). \quad (63)$$

Reducing modulo M and using $(h, M) = 1$ gives $m_{\alpha} \equiv m_{\beta} \pmod{M}$. The inherited row gap (29) then gives $m_{\alpha} = m_{\beta}$. Source separation forces $\alpha = \beta$, a contradiction. $\square$

The lemma removes a local algebraic obstruction, but it does not import a known affine Chowla theorem. The slopes m_{α}, m_{β} and intercepts $h + kM, h + k'M$ grow with X , and the desired estimate sums over row pairs with shared masks. Even a logarithmic saving for every fixed pair would not survive coefficient-blind absolute summation at the required endpoint scale.

5.4 Three distinct saving ledgers

It is useful to distinguish three normalization problems:

object	coherent scale	target scale	required gain
raw equality terminal $k = 0$	$Q^3 J$	Q^3 / J	$J^2 = Q / K_B$
minimal folded or full trace terminal	$Q^3 J K_B$	$Q^2 J K_B$	Q
scalar length- K_B progression	$M K_B^2$	$M K_B$	K_B

The same-row theorem removes the scalar long-gap expression from the minimal folded physical gate. It does not turn the cross-row row-energy problem into the third line of the table.

6 Hilbert-valued one-Möbius compression

The four-factor expansion is exact but can obscure the operator that must be estimated. We now compress each fixed-alias column whose norm square generates the corresponding fixed-alias row form: one target Möbius factor becomes the outer scalar, while all row information is retained in a Hilbert coefficient.

6.1 An exact vector identity

Fix an alias k . Let

$$a_k(r) = \log(r + h + kM) \Phi_0(r + h + kM), \quad (64)$$

interpreted as zero outside the positive terminal support. Define a vector $\mathbf{G}_k^L(r) \in \ell^2\{(\gamma, j)\}$ by

$$[\mathbf{G}_k^L(r)]_{\gamma,j} = \sum_{\alpha: m_{\alpha}j=r} c_{\alpha,\gamma,j}^L a_k(r). \quad (65)$$

For fixed (γ, j, r) , source separation leaves at most one α in this sum.

For comparison with the inherited atomic diagonal, set

$$\mathcal{D}_{k,L} = \sum_{\gamma,j,\alpha} |c_{\alpha,\gamma,j}^L|^2 |\mu(m_{\alpha}j + h + kM)|^2 |a_k(m_{\alpha}j)|^2. \quad (66)$$

Proposition 6.1 (Exact Hilbert compression). *The fixed–alias terminal column satisfies*

$$\boxed{\mathcal{T}_k^L = - \sum_r \mu(r + h + kM) \mathbf{G}_k^L(r).} \quad (67)$$

Moreover, its deterministic coefficient energy is

$$\sum_r \|\mathbf{G}_k^L(r)\|_2^2 = \sum_{\gamma, j, \alpha} |c_{\alpha, \gamma, j}^L|^2 |a_k(m_\alpha j)|^2 =: \tilde{\mathcal{D}}_{k, L}. \quad (68)$$

The actual atomic diagonal $\mathcal{D}_{k, L}$, which includes $|\mu(r + h + kM)|^2$, is at most $\tilde{\mathcal{D}}_{k, L}$; both obey the same deterministic endpoint upper bound. The right column has the identical formulation.

Proof. Insert $r = m_\alpha j$ into (23) and group terms with the same r . This gives (67). When the norm in (65) is squared, different j are different coordinates, and source separation prevents two rows from contributing to the same fixed (j, r) . Hence no cross term appears and (68) follows. $\square$

The equality–alias target can now be stated as a vector Möbius disjointness inequality. Since $\tilde{\mathcal{D}}_{0, L} + \tilde{\mathcal{D}}_{0, R} \ll X^{o(1)} Q^2 J$, a sufficient estimate is

$$\left\| \sum_r \mu(r + h) \mathbf{G}_0^L(r) \right\|_2^2 + \left\| \sum_r \mu(r + h) \mathbf{G}_0^R(r) \right\|_2^2 \ll X^{o(1)} K_B \sum_r (\|\mathbf{G}_0^L(r)\|_2^2 + \|\mathbf{G}_0^R(r)\|_2^2). \quad (69)$$

The coefficient–blind constant is of order Q ; the target constant is only K_B , so the required aggregate row gain is $Q/K_B = X^{o(1)} J^2$.

For the full alias trace, a cleaner but stronger sufficient theorem is

$$\begin{aligned} & \sum_{k=-L}^L \left\| \sum_r \mu(r + h + kM) \mathbf{G}_k^L(r) \right\|_2^2 \\ & + \sum_{k=-L}^L \left\| \sum_r \mu(r + h + kM) \mathbf{G}_k^R(r) \right\|_2^2 \\ & \ll X^{o(1)} \sum_{k, r} (\|\mathbf{G}_k^L(r)\|_2^2 + \|\mathbf{G}_k^R(r)\|_2^2). \end{aligned} \quad (70)$$

This asks for constant–scale fixed–alias row decorrelation after the full alias trace has deleted every cross–alias product.

6.2 Why scalar compression is insufficient

If the physical output coordinates are suppressed, one may formally write a row sum as

$$R_h = \sum_{\ell, d, j} \mu(d) (\log \ell) W(\ell, d, j) \mu(\ell dj + h) = \sum_n G(n) \mu(n + h), \quad (71)$$

where

$$G(n) = \sum_{\ell dj = n} \mu(d) (\log \ell) W(\ell, d, j). \quad (72)$$

This is an identity, not a cancellation estimate. In particular:

- (i) the divisor Möbius factor is stored inside G , not removed;
- (ii) the physical quantity is a sum of squared coordinate outputs, not the square of a scalar sum over coordinates; and

(iii) dualizing the Hilbert norm introduces arbitrary ℓ^2 dual coefficients, so a scalar estimate that cancels between (γ, j) coordinates does not imply (69).

TPC-36 nevertheless supplies meaningful structure. For each fixed opposite row γ , its projective reassembly decomposes the actual coefficient into signed layers of total mass $X^{o(1)}$, each of the form

$$b_\gamma(\ell)f_\gamma(d)\eta_\gamma(j/J), \quad (73)$$

with the ℓ -factor independent of (d, j) . Thus a uniform Hilbert-valued Möbius theorem for this restricted layer class could be reassembled by Minkowski with only an $X^{o(1)}$ loss. The existing projective decomposition does not itself prove that theorem: the common opposite row, gcd mask, row separation, residue cell, and orbit weight remain part of the vector coefficient.

Conditional input 6.2 (Restricted vector Möbius input). For every projective layer generated by TPC-36, uniformly in its opposite row and admissible alias, the corresponding vectors satisfy either (69) or the stronger trace estimate (70).

Conditional input 6.2 is a precise continuation interface, not a result of the present paper. It is coefficient-specific and therefore lies outside the universal positive phase methods ruled out by TPC-40.

7 Raw Möbius decorrelation for triple-prime CRT moduli

We now prove a scalar arithmetic theorem at the exact CRT exponent. The result is independent of the row-diagonal identity, but together they separate what current arithmetic can and cannot estimate.

7.1 The admissible modulus family

Let

$$Y = X^{133/400}, \quad M_0 = Y^3 = X^{399/400}, \quad K = X/M_0 = X^{1/400}. \quad (74)$$

Let $C_{\mathcal{J}}$ be a fixed constant with $\max \mathcal{J} \leq C_{\mathcal{J}}Y$. Choose constants

$$C_{\mathcal{J}} < a_1 < b_1 < a_2 < b_2 < a_3 < b_3 \quad (75)$$

and the three pairwise disjoint fixed-proportion intervals

$$I_i(X) = [a_i Y, b_i Y], \quad i = 1, 2, 3, \quad (76)$$

The inherited source exponents give $\max d_\alpha = o(Y)$ and $Y = o(\min \ell_\alpha)$, while $|h| = O(1)$. Thus every prime in these intervals satisfies (15) for sufficiently large X . The constants are fixed and the intervals have positive relative length. Put

$$\mathcal{P}_i = I_i(X) \cap \{\text{primes}\}, \quad \mathfrak{M}_X = \{q_1 q_2 q_3 : q_i \in \mathcal{P}_i\}. \quad (77)$$

Disjointness makes this parametrization injective. The prime number theorem gives

$$|\mathfrak{M}_X| \asymp \frac{Y^3}{(\log Y)^3} \asymp \frac{M_0}{(\log X)^3}. \quad (78)$$

Every $M \in \mathfrak{M}_X$ is squarefree, $M \asymp M_0$, $\varphi(M) \asymp M$, and

$$K_M := X/M \asymp K. \quad (79)$$

Fix a constant C_0 larger than the maximum product of the interval endpoints and put

$$Q_* = C_0 M_0. \quad (80)$$

Then every member of $\mathfrak{M}_X$ is at most Q_* , and $Q_* \leq x/10$ uniformly for $x \asymp X$ and large X .

For $x > 0$, define the reduced residue vector and its raw energy by

$$S_M(a; x) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{M}}} \mu(n), \quad (81)$$

$$V_M(x) = \sum_{a \bmod M}^* |S_M(a; x)|^2. \quad (82)$$

Theorem 7.1 (Raw triple–prime progression decorrelation). *Fix*

$$0 < \theta < \frac{1}{200}. \quad (83)$$

Then, for all but $o(|\mathfrak{M}_X|)$ moduli $M \in \mathfrak{M}_X$,

$$\boxed{V_M(X) \ll_{\theta} \frac{X K_M}{(\log X)^{\theta}}.} \quad (84)$$

The sum is raw and uncentered: no Dirichlet character main term remains. More quantitatively, the exceptional proportion is

$$\ll (\log X)^3 \exp(-c(\log X)^{1-200\theta}) + \frac{(\log X)^{C_1+\theta}}{K}, \quad (85)$$

where $c > 0$ and C_1 are absolute.

The trivial residue–class bound is

$$V_M(X) \leq \varphi(M)(K_M + 1)^2 \ll X K_M. \quad (86)$$

Thus (84) is a coefficient–specific logarithmic saving.

7.2 The centered almost–all theorem

Set

$$\varepsilon = (\log X)^{-\theta}. \quad (87)$$

We use Klurman et al. [3, Theorem 1.4]. In the present notation, their result supplies a good set of moduli such that for every good, typical M there is a Dirichlet character $\chi_M \pmod{M}$ for which

$$V_M^{\circ}(x) := \sum_{a \bmod M}^* \left| S_M(a; x) - \frac{\chi_M(a)}{\varphi(M)} T_M(x) \right|^2 \ll \varepsilon \varphi(M) (x/M)^2, \quad (88)$$

where, after replacing χ_M by its conjugate if necessary,

$$T_M(x) = \sum_{n \leq x} \mu(n) \overline{\chi_M(n)}. \quad (89)$$

Take the maximum modulus parameter in that theorem to be Q_* from (80). Since $x/Q_* \asymp K$ for $x \asymp X$, the condition

$$(\log(x/Q_*))^{-1/200} \leq \varepsilon \quad (90)$$

holds for large X precisely because $\theta < 1/200$. The number of exceptional integer moduli is at most

$$\ll Q_* x^{-\varepsilon^{200}} \ll M_0 \exp(-c(\log X)^{1-200\theta}). \quad (91)$$

By (78), its proportion inside $\mathfrak{M}_X$ is

$$\ll (\log X)^3 \exp(-c(\log X)^{1-200\theta}) = o(1). \quad (92)$$

For completeness, the typicality condition is automatic here. Its small-prime threshold is

$$y = (x/Q_*)^{\varepsilon^2} = X^{o(1)}, \quad (93)$$

whereas every prime factor of M is comparable with $Y = X^{133/400}$. There are no prime divisors below the first such factor, and above it there are at most three, while $\pi(z)/100 > 3$ for large X . Hence every member of $\mathfrak{M}_X$ is typical in the sense required by the cited theorem.

It remains to dispose of the rank-one term in (88). This is not automatic from the centered theorem and is the main additional argument.

7.3 A maximal multiplicative large sieve

We record the standard maximal form needed below.

Lemma 7.2 (Maximal multiplicative large sieve). *For complex coefficients a_n , $R, Z \geq 2$,*

$$\sum_{r \leq R} \frac{r}{\varphi(r)} \sum_{\psi \bmod r}^* \max_{z \leq Z} \left| \sum_{n \leq z} a_n \psi(n) \right|^2 \ll (Z + R^2)(\log(2Z))^2 \sum_{n \leq Z} |a_n|^2. \quad (94)$$

The star denotes primitive characters.

Proof. The ordinary multiplicative large sieve gives the same inequality without the maximum and logarithmic factor; see, for example, Iwaniec and Kowalski [2, Chapter 7]. Decompose every initial interval $[1, z]$ into at most $O(\log Z)$ dyadic blocks. Cauchy–Schwarz costs one logarithm. Apply the ordinary large sieve to each block and sum over all dyadic scales; every coefficient belongs to one block at each scale, costing the second logarithm. This proves (94). $\square$

7.4 Deleting the induced character on average

The following lemma applies to an arbitrary assignment of one character to each modulus, so in particular to the characters selected in (88). The principal conductor is separated because its classical bound is individually sufficient at the target scale, even though it need not be as small as the nonprincipal family average.

Lemma 7.3 (One-character disposal). *For each $M \in \mathfrak{M}_X$, choose any character $\chi_M \pmod{M}$, and let r_M be the conductor of its primitive inducing character. Uniformly for $x \asymp X$,*

$$\boxed{\frac{1}{|\mathfrak{M}_X|} \sum_{\substack{M \in \mathfrak{M}_X \\ r_M > 1}} \frac{1}{\varphi(M)} \left| \sum_{n \leq x} \mu(n) \overline{\chi_M(n)} \right|^2} \ll X(\log X)^C \quad (95)$$

for an absolute constant C . If $r_M = 1$, then individually

$$\frac{|T_M(x)|^2}{\varphi(M)} \ll XK \exp\left(-c(\log K)^{3/5}(\log \log K)^{-1/5}\right). \quad (96)$$

Proof. Let ψ_M be the primitive character inducing χ_M , write

$$r_M = \text{cond}(\psi_M) \mid M, \quad s_M = M/r_M, \quad t = \omega(r_M) \in \{0, 1, 2, 3\}. \quad (97)$$

Because M is squarefree, r_M is the product of a subset of its three prime factors.

Put

$$U_\psi(z) = \sum_{n \leq z} \mu(n) \overline{\psi(n)}. \quad (98)$$

Deleting from the Euler product the primes dividing s_M gives the exact finite identity

$$T_M(x) = \sum_{\substack{d \leq x \\ p \mid d \Rightarrow p \mid s_M}} \overline{\psi_M(d)} U_{\psi_M}(x/d). \quad (99)$$

Indeed, the Dirichlet series for the left side is the series for $\mu \overline{\psi}_M$ multiplied by $\prod_{p \mid s_M} (1 - \overline{\psi}_M(p) p^{-z})^{-1}$. Thus d in (99) may contain prime powers, but only primes from s_M . Since every such prime is $\asymp Y$ and $X = Y^{400/133} < Y^{3.01}$, every retained d has $\Omega(d) \leq 3$; the sum consequently has $O(1)$ terms.

Fix $t \geq 1$. A conductor made from t of the three prime intervals can be extended to a member of $\mathfrak{M}_X$ in at most

$$W_t \ll \left(\frac{Y}{\log Y} \right)^{3-t} \quad (100)$$

ways. Since $\varphi(M) \asymp Y^3$, the finite expansion, the maximal large sieve with $a_n = \mu(n)$, and $\sum_{n \leq x} |\mu(n)|^2 \leq x$ give

$$\begin{aligned} \Sigma_t(x) &:= \sum_{\substack{M \in \mathfrak{M}_X \\ \omega(r_M) = t}} \frac{|T_M(x)|^2}{\varphi(M)} \\ &\ll \frac{W_t}{Y^3} (X + Y^{2t}) X (\log X)^C. \end{aligned} \quad (101)$$

Here the large-sieve parameters are $R \asymp Y^t$ and $Z \asymp X$, and $r/\varphi(r) \asymp 1$ on these conductor families. The standard Y^{2t} large-sieve term is retained here. Dividing by (78) yields

$$\frac{\Sigma_t(x)}{|\mathfrak{M}_X|} \ll \frac{X(X + Y^{2t})}{Y^{t+3}} (\log X)^{t+C}. \quad (102)$$

Thus the layers $t = 3, 2, 1$ are respectively

$$\ll X (\log X)^{3+C}, \quad \ll \frac{X}{Y} (\log X)^{2+C}, \quad \ll \frac{X^2}{Y^4} (\log X)^{1+C}. \quad (103)$$

The last two are $o(X)$. Notice especially that for $t = 3$ the large sieve first gives $Y^6 X = M_0^2 X$; division by $\varphi(M) \asymp M_0$ and then by $|\mathfrak{M}_X| \asymp M_0/(\log X)^3$ gives exactly $X (\log X)^{3+C}$. No M_0^2 term has been discarded.

If $t = 0$, χ_M is principal. The same Euler deletion gives

$$T_M(x) = \sum_{\substack{d \leq x \\ p \mid d \Rightarrow p \mid M}} \mathbf{M}(x/d), \quad \mathbf{M}(z) = \sum_{n \leq z} \mu(n). \quad (104)$$

There are $O(1)$ terms and, for every one, $x/d \gg K$. The classical zero-free-region bound [2, Chapter 5]

$$\mathbf{M}(z) \ll z \exp \left(-c (\log z)^{3/5} (\log \log z)^{-1/5} \right) \quad (105)$$

therefore gives (96); after squaring we weaken the exponential by adjusting c . The layers $t = 1, 2, 3$ prove (95). $\square$

7.5 Proof of the raw theorem

Proof of theorem 7.1. For the character selected by (88), the rank-one vector is the orthogonal projection of $(S_M(a; X))_{a \in (\mathbb{Z}/M\mathbb{Z})^\times}$ onto χ_M . Hence the Pythagorean identity is exact:

$$V_M(X) = V_M^\circ(X) + \frac{|T_M(X)|^2}{\varphi(M)}. \quad (106)$$

Outside the exceptional set in (92), the first term is $\ll \varepsilon X K_M$.

If the selected character is principal, then (96) is $o(\varepsilon X K_M)$, uniformly in M . On the remaining moduli, lemma 7.3 and Markov's inequality show that the proportion for which the second term exceeds $\varepsilon X K_M$ is

$$\ll \frac{(\log X)^C}{\varepsilon K} = o(1), \quad (107)$$

because $K = X^{1/400}$. Together with (92), this gives (85). Delete this set and use (106). Since $\varepsilon = (\log X)^{-\theta}$, the result is (84). $\square$

Corollary 7.4 (RMS cancellation and moving-shift aggregation). *For every modulus supplied by theorem 7.1,*

$$\left(\frac{1}{\varphi(M)} \sum_{a \bmod M}^* |S_M(a; X)|^2 \right)^{1/2} \ll \frac{K_M}{(\log X)^{\theta/2}}. \quad (108)$$

Moreover,

$$\begin{aligned} V_M(X) &= \sum_{\substack{n \leq X \\ (n, M)=1}} \mu(n)^2 \\ &\quad + 2 \sum_{r \geq 1} \sum_{\substack{n \leq X-rM \\ (n, M)=1}} \mu(n) \mu(n+rM). \end{aligned} \quad (109)$$

Thus (84) gives a logarithmic saving for the signed aggregate of all M -spaced scalar correlations.

Proof. The RMS estimate follows from $\varphi(M) \asymp M = X/K_M$. Expanding the raw square, writing two congruent integers as n and $n+rM$, and separating the diagonal proves (109). $\square$

The word “aggregate” is essential. The theorem does not bound the absolute value of each fixed- r correlation, nor does it insert the physical row weights of (56).

8 Bounded-variation transfer to terminal weights

The terminal atom contains a smooth cutoff and a logarithm. We extend the raw prefix theorem to such weights without invoking a separate smooth version of the short-progression theorem.

Let $0 < c < C < \infty$ be fixed. For a function W supported on $[c, C]$, write

$$\|W\|_{\text{BV}} := \|W\|_\infty + \text{Var}_{[c, C]}(W). \quad (110)$$

The proof of theorem 7.1 is uniform for $x \asymp X$. Below, $e_{n \bmod M}$ denotes the standard basis vector when $(n, M) = 1$ and the zero vector otherwise.

Theorem 8.1 (Weighted raw progression theorem). *Under the hypotheses of theorem 7.1, let W_X be any family supported in $[c, C]$ with $\|W_X\|_{\text{BV}} \ll 1$. Then, for all but $o(|\mathfrak{M}_X|)$ moduli,*

$$\left| \sum_{a \bmod M}^* \sum_{\substack{n \equiv a \\ (\bmod M)}} \mu(n) W_X(n/X) \right|^2 \ll \frac{X K_M}{(\log X)^\theta}. \quad (111)$$

The exceptional set may be chosen uniformly for any fixed finite collection of such weights.

Proof. Retain $\varepsilon = (\log X)^{-\theta}$ and choose

$$R = \lceil \varepsilon^{-1/2} \rceil. \quad (112)$$

Take an R -point grid on $[cX, CX]$. Apply the proof of [theorem 7.1](#) at every grid endpoint. The union of the KMT exceptional sets is still zero density, since the extra factor R is logarithmic and (92) decays faster than every logarithm. Summing the character energies over the grid and applying Markov costs the same factor R , but

$$\frac{R(\log X)^C}{\varepsilon K} = o(1). \quad (113)$$

Thus, outside a zero-density set, every grid prefix vector has squared norm $\ll \varepsilon X K_M$.

Between adjacent endpoints, an interval has length $O(X/R)$, so each residue class contains at most $O(K_M/R + 1)$ integers. Its coefficient-blind residue energy is therefore

$$\ll M(K_M/R + 1)^2 \ll \frac{X K_M}{R^2} \ll \varepsilon X K_M, \quad (114)$$

because $K_M/R \rightarrow \infty$. The raw prefix vector consequently has norm $O((\varepsilon X K_M)^{1/2})$ uniformly throughout $[cX, CX]$.

Abel summation in the Hilbert space $\ell^2((\mathbb{Z}/M\mathbb{Z})^\times)$ now gives

$$\left\| \sum_n \mu(n) W_X(n/X) e_{n \bmod M} \right\|_2 \leq \|W_X\|_{\text{BV}} \sup_{cX \leq t \leq CX} \left\| \sum_{n \leq t} \mu(n) e_{n \bmod M} \right\|_2. \quad (115)$$

Squaring proves (111). $\square$

Let Φ be a fixed smooth function supported on a compact subinterval of $(0, \infty)$, and put

$$W_X(t) = \frac{\log(Xt)}{\log X} \Phi(t). \quad (116)$$

Its bounded-variation norm is uniform. Therefore [theorem 8.1](#) gives

$$\sum_{a \bmod M}^* \left| \sum_{n \equiv a \pmod{M}} \mu(n) \log n \Phi(n/X) \right|^2 \ll \frac{X K_M (\log X)^2}{(\log X)^\theta}. \quad (117)$$

This matches the scalar shape of the terminal atom (22) after a harmless normalization of its fixed cutoff.

Remark 8.2 (What transfers, and what does not). The theorem controls the full reduced residue vector with a common scalar weight. Any unweighted subset of residue coordinates inherits the bound by positivity. The actual TPC output, however, assigns row-dependent coefficients to a Hilbert coordinate indexed by the opposite row and orbit time. Neither coordinatewise restriction nor (117) supplies the vector inequality (69).

9 The completed convolution shadow and its boundary defect

There is a second reason not to attack the expanded four-Möbius form term by term. In an idealized scalar completion, the divisor-row coefficient collapses exactly under Dirichlet convolution.

Let $*$ denote Dirichlet convolution, let Λ be the von Mangoldt function, and let $\mathbf{1}(n) = 1$.

Proposition 9.1 (Full divisor completion). *For every $n \geq 1$,*

$$\boxed{(\Lambda * \mu * \mathbf{1})(n) = \Lambda(n)}. \quad (118)$$

Consequently, for every finitely supported scalar weight depending only on the product $n = \ell dj$, the fully completed model satisfies

$$\sum_{\ell, d, j} \Lambda(\ell) \mu(d) \mu(\ell dj + h) W(\ell dj) = \sum_n \Lambda(n) \mu(n + h) W(n). \quad (119)$$

Proof. Möbius inversion gives $\mu * \mathbf{1} = \delta_1$, the identity for Dirichlet convolution. Associativity gives $\Lambda * (\mu * \mathbf{1}) = \Lambda$. Grouping the finite triple sum by $n = \ell dj$ proves (119). $\square$

The right side of (119) is the shifted prime–Möbius correlation studied on average over h by Lichtman [4] and in the broader Hardy–Littlewood–Chowla setting by Lichtman and Teräväinen [5]. In particular, known results give qualitative cancellation for almost all shifts once the shift range is sufficiently larger than a power of $\log X$.

This identity does not apply unchanged to the physical packet. A single projective layer contains a truncated coefficient of the shape

$$\sum_{\ell dj=n} b_\gamma(\ell) f_\gamma(d) \eta_\gamma(j/J), \quad (120)$$

where the factors are restricted to prescribed dyadic ranges and inherit the opposite–row mask. Even before that mask, the natural model is a localized convolution

$$\Lambda_{L_0} * \mu_{D_0} * \mathbf{1}_{\mathcal{J}}, \quad (121)$$

where the subscripts denote restriction to the inherited source dyadic scales L_0, D_0 and orbit interval $\mathcal{J}$. It is not $\Lambda * \mu * \mathbf{1}$. The difference between (121) and an appropriate localized Λ term is a boundary defect whose Hilbert norm has not been estimated here.

Two upgrades would be required to turn the identity into a physical theorem:

- (i) a localization theorem controlling the dyadic and mask boundary defect with power strong enough for (69); and
- (ii) a vector–valued shifted–prime estimate uniform in the opposite row and projective layer, rather than an average over the external shift h .

The exact convolution collapse is therefore a route marker. It shows that part of the apparent fourfold structure is created by truncation, but writing the completion is not itself a bound for the truncation error.

There is one further asymmetry. At alias $k = 0$, $(d_\alpha, m_\alpha j + h) = 1$, so the two Möbius values can sometimes be fused multiplicatively. At $k \neq 0$, the condition $(d_\alpha, h + kM) = 1$ is not automatic. Any fusion argument must either remain on the equality alias or explicitly control these new gcd faces.

10 Relation to existing Möbius correlation theorems

The scalar theorem in section 7 uses a result designed for large moduli. Other major advances on Möbius correlations illuminate the boundary but do not directly estimate the physical form.

10.1 Short arithmetic progressions

The theorem of Klurman et al. [3] gives, outside a nearly power-saving exceptional set of moduli, centered variance

$$\ll \varepsilon \varphi(M)(X/M)^2. \quad (122)$$

At the TPC scale this is εXK . Our lemma 7.3 removes their rank-one term for almost all members of the declared triple-prime family, so that the raw Möbius vector has the same logarithmic saving.

The scalar saving by itself is not new. The classical Barban–Davenport–Halberstam theorem for μ , as recalled in Klurman et al. [3, Section 3.1.3], gives for every fixed $A > 0$, with a suitable $B = B(A)$,

$$\sum_{q \leq X/(\log X)^B} \sum_{a \bmod q}^* \left| \sum_{\substack{n \leq X \\ n \equiv a \pmod{q}}} \mu(n) \right|^2 \ll_A \frac{X^2}{(\log X)^A}. \quad (123)$$

Because $M_0 = X^{399/400}$ eventually lies below every such cutoff, restriction to $\mathfrak{M}_X$ and Markov’s inequality already give arbitrary fixed logarithmic savings outside a logarithmic-density exception. The additional content of our KMT-large-sieve route is the explicit stronger exceptional proportion (85), together with the admissible three-interval and bounded-variation terminal specialization.

The result remains above the scalar atomic diagonal:

$$\frac{\varepsilon XK}{X} = \varepsilon K = \frac{X^{1/400}}{(\log X)^\theta} \longrightarrow \infty. \quad (124)$$

Thus it is not a diagonal-scale progression theorem. Assuming GRH can remove or reduce exceptional-modulus issues in the cited framework, but it does not by itself supply the missing factor $K/(\log X)^\theta$ in (124).

10.2 Averaged Chowla and fixed affine forms

The averaged Chowla theorem of Matomäki et al. [6] gains cancellation after averaging over all shifts in a dense box. The alias differences here are the sparse multiples rM , and the physical cross-row kernel also has growing slopes m_α, m_β . Restricting a dense shift average to that sparse subset loses the density factor that one is trying to save.

Results at almost all scales for fixed shifts [8] and logarithmically averaged two-point results [7] likewise do not give a uniform estimate for (62) when all coefficients vary with X . The Bohr-set analogue of Teräväinen and Walker [9] has fixed Bohr data and logarithmic averaging; the TPC modulus, frequencies, row slopes, and masks all move.

10.3 Shifted primes

The ideal convolution shadow in (119) lies within the averaged-shift theory of Lichtman [4], Lichtman and Teräväinen [5]. This is a genuine positive indication for the completion route. It does not cover a fixed twin shift with the actual localized vector weights. Averaging over h would change the problem: it may prove an almost-all-shift theorem, but it cannot by itself close the prescribed fixed- h physical packet.

10.4 The parity boundary

The surviving form contains an even number of Möbius factors and is attached to a prime-pair sieve endpoint. This is consistent with the classical parity obstruction [1]. The present paper

makes two limited advances:

- (i) it proves that the same-row two-Möbius alias sector was a Gram artifact and removes it unconditionally; and
- (ii) it obtains a logarithmic saving for the raw scalar CRT residue vector on almost all admissible triple-prime moduli.

Neither statement estimates the cross-row form (56). We therefore do not describe either one as a breach of parity.

11 What has closed and what remains

The endpoint can now be stated with fewer moving parts.

11.1 Unconditional advances

The following statements are proved without a new conjectural input:

- (i) the complete signed same-row unequal-alias sector in the minimal bank is bounded by the existing atomic diagonal;
- (ii) the full alias DFT deletes that sector exactly;
- (iii) the physical terminal bank is reduced to the cross-row form $\mathcal{F}_L + \mathcal{F}_R$;
- (iv) every affine target pair in that form is algebraically nonproportional;
- (v) the form admits an exact Hilbert-valued one-Möbius compression; and
- (vi) almost all admissible triple-prime CRT moduli give a raw scalar progression energy smaller than the coherent scale by $(\log X)^\theta$, including the smooth terminal weight.

The sixth item is a useful quantitative specialization with a stronger exceptional-density ledger than the elementary classical benchmark. It should not be read as the first known scalar logarithmic saving in this range. The first five items remove false alternatives and specify the coefficient class that a stronger theorem must address.

11.2 The next analytic theorem

The exact continuation for the selected minimal folded route is the one-sided estimate

$$\operatorname{Re}(\mathcal{F}_L + \mathcal{F}_R) \ll X^{o(1)} Q^2 J K_B. \quad (125)$$

Alternatively, the original equality column may be attacked directly through the distinct sufficient vector inequality (69), without first bounding the entire folded energy. A stronger, cleaner target is the full trace inequality (70).

There are three concrete routes, with distinct failure modes:

- (a) **Vectorize the progression theorem.** Prove a Bessel or large-sieve inequality for the actual TPC-36 projective coefficient class. A scalar residue variance is insufficient unless the upgrade is uniform under arbitrary ℓ^2 dualization of the output coordinates.
- (b) **Control the localized convolution defect.** Compare $\Lambda_{L_0} * \mu_{D_0} * \mathbf{1}_{\mathcal{J}}$, with its opposite-row masks, to a localized Λ model in a Hilbert norm. The full identity $\Lambda * \mu * \mathbf{1} = \Lambda$ identifies the hoped-for main term but supplies no boundary estimate.

- (c) **Average the selectable CRT triple.** The auxiliary primes are chosen inside admissible intervals. A nonnegative average of the *complete physical vector energy* over those triples would select at least one good triple. The scalar theorem of [theorem 7.1](#) shows that this family is arithmetically useful, but an average of scalar residue energies is not yet an average of $\mathcal{F}_L + \mathcal{F}_R$.

The routes may be combined: a triple average could first yield a vector bound for most projective layers, while a convolution estimate treats the exceptional masks. Conversely, a rigorous obstruction showing that one route cannot reach (125) would itself delimit the method class and justify returning to another branch.

11.3 Claim boundary

This paper does *not* prove:

- (i) cancellation for an individual long shift rM ;
- (ii) a diagonal-scale scalar progression variance;
- (iii) the Hilbert-valued physical estimate (69);
- (iv) cancellation in the cross-row four-Möbius form;
- (v) a breach of sieve parity;
- (vi) a Hardy-Littlewood prime-pair asymptotic; or
- (vii) infinitely many twin primes.

The result is a forward reduction plus a strict scalar logarithmic saving. The remaining distance is explicitly the factor $K/(\log X)^\theta$ in the scalar shadow and the scalar-to-physical Hilbert upgrade in the true row gate.

12 Exact certificate and claim ledger

The proofs above are analytic and exact. A companion standard-library certificate checks the finite cyclic and convolution identities using integers, rational arithmetic, modular character orthogonality, and formal prime-logarithm vectors. It uses no floating point, random draws, or optimization-sensitive `assert` statements.

12.1 Certified identities

The frozen certificate verifies:

- (i) the minimal alias Gram, its 2, 1, 0 spectrum and folded bucket decomposition for a declared range of L ;
- (ii) the full $(2L + 1)$ -point DFT identity $H = I$;
- (iii) exact same-row energy decompositions and the sharp inequality $|\mathcal{S}_{\text{same}}| \leq \mathcal{D}$ on exhaustive small integer atom families;
- (iv) the allowed alias pairs in (60);
- (v) the nonproportional determinant implication on exhaustive finite admissible row-modulus models;
- (vi) the formal convolution identity $\Lambda * \mu * \mathbf{1} = \Lambda$ through the frozen integer range; and

(vii) all endpoint exponent identities, including

$$3\frac{133}{400} = \frac{399}{400}, \quad 1 - \frac{399}{400} = \frac{1}{400}, \quad 2\frac{267}{400} + \frac{133}{400} + \frac{1}{400} = \frac{167}{100}. \quad (126)$$

Running

```
python experiments/tpc41_certificate.py
python -O experiments/tpc41_certificate.py
```

produces byte-identical canonical JSON records. The frozen run passes

$$\boxed{4,358,451 \text{ exact checks.}} \quad (127)$$

It includes 80 minimal and 80 full alias Grams, 81,375 real and 3,250 Gaussian-integer atom vectors, 80,400 allowed ordered alias pairs, 693,840 affine determinants, and the formal convolution through $n = 10,000$. The certificate digest is

e10174e2448d5323afb586d5855112e596ea7e134cd5899f216ef3a916748fc4.

The SHA-256 hashes of the canonical JSON record and source are, respectively,

a12acada9747cb53acf1a5bdc7c763ede4ace5254cc7ce084145c596ada4c107,
b08c52e25f8c59105017cb9e602039e7030d4671456be9eb77a22d36b84f54f0.

12.2 What the certificate does not certify

The computation does not verify the asymptotic theorem of Klurman et al. [3], the prime number theorem, the Mertens zero-free-region estimate, or the multiplicative large sieve; these are cited analytic inputs. It is not numerical evidence for random Möbius signs. Its JSON claim ledger explicitly marks as false any claim that it proves the physical four-Möbius estimate, a parity breach, a prime-pair asymptotic, or the twin-prime conjecture.

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